

DRUMALBIN, United KingdomWMO: **031550**Lat: **55.6272N**Lon: **3.7361W**Elev: **804**StdP: **14.27**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.4	25.6	20.0	15.5	25.2	22.5	17.4	28.2	42.8	42.4	39.5	42.1	3.6	100	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	71.1	62.0	67.9	60.0	65.1	58.6	63.5	68.9	61.6	66.0	59.9	63.7	10.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.3	83.4	65.6	59.6	78.6	63.6	58.1	74.3	62.0	29.2	68.7	27.8	66.1	26.7	63.7	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.8	30.6	27.1	17.8	77.1	4.3	3.5	14.6	79.6	12.1	81.7	9.7	83.6	6.5	86.2		
(5)			17.5	67.3	4.2	3.3	14.4	69.6	11.9	71.5	9.5	73.4	6.4	75.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.3	37.4	37.5	40.0	44.2	48.7	53.5	56.8	56.2	53.1	47.6	41.7	37.8		
	DBStd	8.66	5.38	5.02	4.91	4.84	5.21	4.32	4.04	3.65	4.14	4.87	5.49	6.66		
	HDD50	2080	390	349	310	182	86	13	1	1	16	102	251	379		
	HDD65	6839	855	769	775	623	504	344	258	273	358	539	698	843		
	CDD50	718	0	0	0	9	47	119	210	194	108	28	3	0		
	CDD65	4	0	0	0	0	0	0	3	1	0	0	0	0		
	CDH74	37	0	0	0	0	1	8	20	8	0	0	0	0		
	CDH80	2	0	0	0	0	0	1	1	0	0	0	0	0		
	WSAvg	12.4	14.4	14.3	14.0	12.5	11.8	10.9	9.8	10.5	11.8	12.7	12.7	13.3		
	PrecAvg	46.7	5.3	4.1	3.5	2.7	2.9	2.8	3.5	3.6	3.5	5.0	4.8	5.2		
(15) Precipitation	PrecMax	56.9	9.8	10.5	7.1	4.2	5.1	6.7	5.9	9.0	6.0	8.6	10.0	12.2		
	PrecMin	34.4	0.7	0.7	0.9	0.6	0.6	1.0	1.1	0.7	0.7	1.3	2.0	1.2		
	PrecStd	6.1	2.1	2.5	1.3	0.9	1.3	1.3	1.3	2.0	1.4	2.2	2.1	2.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.2	51.4	56.8	65.2	71.1	74.4	76.8	73.6	68.7	60.9	56.0	53.0		
		MCWB	49.3	48.9	48.2	55.1	60.0	62.3	64.4	64.7	60.8	57.0	54.1	51.5		
	2%	DB	48.9	48.8	51.4	58.8	66.5	68.8	71.7	68.0	64.7	58.3	53.4	50.8		
		MCWB	47.6	46.8	47.3	50.7	56.9	60.4	63.1	61.2	58.5	55.6	51.8	49.2		
	5%	DB	47.1	47.1	49.1	54.9	61.6	65.0	68.0	65.2	62.1	56.6	51.5	49.0		
		MCWB	45.6	45.3	45.9	49.2	54.3	58.2	60.5	59.4	57.5	54.3	49.9	47.5		
	10%	DB	45.4	45.3	47.3	52.0	57.6	61.8	64.6	62.9	59.9	54.9	49.8	46.9		
		MCWB	44.0	43.4	44.6	47.5	51.6	56.2	58.8	58.2	56.1	52.6	48.0	45.4		
	0.4%	WB	49.9	50.0	50.4	55.4	62.0	64.8	66.6	65.2	63.0	58.4	54.6	51.7		
		MCDB	50.6	50.8	54.3	63.0	69.7	71.5	74.4	70.9	66.8	59.4	55.5	52.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.9	47.4	48.3	52.3	57.9	61.7	63.9	62.4	60.2	56.3	52.1	49.5		
		MCDB	48.8	48.5	50.6	57.3	64.5	67.3	70.0	66.0	62.8	57.7	53.1	50.6		
	5%	WB	45.9	45.6	46.6	50.0	55.1	59.0	61.6	60.8	58.5	54.7	50.3	47.7		
		MCDB	46.9	46.9	48.4	53.8	60.5	63.7	66.4	64.1	61.0	56.3	51.5	48.9		
	10%	WB	44.1	43.7	45.0	48.2	52.7	56.8	59.6	59.2	56.7	52.9	48.3	45.6		
		MCDB	45.3	45.2	47.0	51.4	56.9	60.8	63.6	62.1	59.3	54.7	49.6	46.9		
	5% DB	MDBR	7.6	8.8	10.6	12.8	14.3	13.5	13.3	12.7	11.5	10.0	8.6	8.1		
		MCDBR	8.9	9.7	12.7	17.5	20.5	19.3	19.2	15.6	14.4	10.6	9.3	9.5		
		MCWBR	8.9	9.0	9.5	11.7	12.7	12.3	11.6	9.8	9.6	8.8	9.0	9.5		
		MCWBR	8.9	9.3	11.7	15.0	18.5	17.7	16.7	13.8	12.2	9.9	9.6	9.6		
(35) Mean Daily Temperature Range	5% WB	MCWBR	9.0	9.1	9.6	10.8	12.2	12.4	11.0	9.9	9.5	9.1	9.5	9.6		
	taub		0.316	0.328	0.354	0.373	0.375	0.372	0.373	0.358	0.347	0.331	0.308	0.299		
		taud	2.236	2.344	2.308	2.285	2.321	2.354	2.354	2.407	2.447	2.436	2.325	2.212		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		185	235	256	268	275	277	273	271	258	232	190	160		
		Edh at Noon	18	24	31	36	37	36	36	32	27	21	16	15		
	RadAvg		167	371	705	1144	1492	1451	1396	1128	803	441	217	124		
(45) All-Sky Solar Radiation	RadStd		24	38	71	117	117	144	149	88	64	55	21	12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page